

AI and ML in Daily Life

Introduction to Artificial Intelligence

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Objective

In this activity, five real-world examples of Artificial Intelligence (AI) and Machine Learning (ML) encountered in daily life are identified. For each example, three questions are answered:

- What does it do?
- What data does it use?
- Is it Supervised, Unsupervised, or Reinforcement Learning?

Example 1: Smartphone Face Unlock

- **What it does:** When you try to unlock your phone, the front camera scans your face and compares it to a stored model of your face. If they match, the phone unlocks. This happens in under a second.
- **Data it uses:** Thousands of labelled photos of your face taken during setup, facial geometry data (such as the distance between your eyes, nose, and mouth), and infrared depth maps.
- **Type of ML:** Supervised Learning
- **Explanation:** The model was trained on a large dataset of labelled face images. It learned to distinguish the phone owner's face from all other faces. Because the training data has clear labels (owner vs. not owner), this is supervised learning.

Example 2: Streaming Recommendations (Netflix / Showmax)

- **What it does:** After you watch a few films or series, the platform suggests new content it thinks you will enjoy. The more you watch, the more accurate the suggestions become.
- **Data it uses:** Your watch history, the ratings you give, how long you watch before stopping, the time of day you watch, your search queries, and anonymous viewing data from millions of other users.

- **Type of ML:** Unsupervised Learning (Collaborative Filtering)
- **Explanation:** The system groups users with similar taste without anyone manually labelling the groups. It then recommends what similar users enjoyed. Because no one tells the algorithm what each group means, this is unsupervised learning.

Example 3: Fraud Detection in Nigerian Banks

- **What it does:** When you make a payment with your debit card or transfer money via a mobile banking app, an AI system checks the transaction in real time. If something looks suspicious, it blocks the transaction or sends an OTP challenge to verify it is really you.
- **Data it uses:** Your transaction history (amounts, merchants, locations, times), your usual login device and location, and anonymized fraud reports from millions of accounts across the bank.
- **Type of ML:** Supervised Learning combined with Unsupervised Learning
- **Explanation:** A supervised model is trained on past fraud cases to recognise known fraud patterns. An unsupervised anomaly detection model runs alongside it to catch completely new fraud types that have never been seen before. Nigerian banks such as GTBank and Access Bank use systems like this.

Example 4: AI Crop Disease Detection for Nigerian Farmers

- **What it does:** A farmer takes a photo of a sick cassava or tomato leaf using a smartphone. An app analyses the photo and tells the farmer what disease the plant has and what treatment to apply. This works even in areas with poor internet.
- **Data it uses:** Thousands of labelled photographs of healthy and diseased crop leaves (labelled by agronomists), soil sensor readings, local weather data (rainfall, temperature, humidity), and GPS location to track regional disease outbreaks.
- **Type of ML:** Supervised Learning
- **Explanation:** A Convolutional Neural Network (CNN) is trained on a large set of crop images, each labelled with the correct disease or health status. Because the training examples are labelled, this is supervised learning. The model learns to classify new photos it has never seen before.

Example 5: Keyboard Autocorrect and Predictive Text

- **What it does:** As you type on your phone, the keyboard fixes spelling mistakes automatically and suggests the next word you are likely to type. Over time it learns your personal vocabulary, including names, slang, and Pidgin English.
- **Data it uses:** large text datasets (books, websites, messages) used to build the initial language model, plus your personal typing history stored on-device, and the corrections you accept or reject.
- **Type of ML:** Supervised Learning and Reinforcement Learning
- **Explanation:** The base language model is trained in a supervised way on correct text. Reinforcement learning then personalizes it over time. Every time you accept a suggested word, that is a positive reward. Every time you delete it and type something else, that is a negative signal. The model updates itself accordingly.

Summary Table

No.	Example	Data Used	ML Type
1.	Phone Face Unlock	Face images, geometry	Supervised
2.	Streaming Recommendations	Watch history, ratings	Unsupervised
3.	Bank Fraud Detection	Transactions, login data	Supervised + Unsupervised
4.	AI Crop Disease Detection	Leaf photos, weather, soil	Supervised
5.	Keyboard Autocorrect	Text corpora, typing history	Supervised + Reinforcement